Swiss TPH

Quarto @ Swiss TPH

Reproducible documents for your research projects

Zhihan Zhu, Research Informatics

28.10.2026

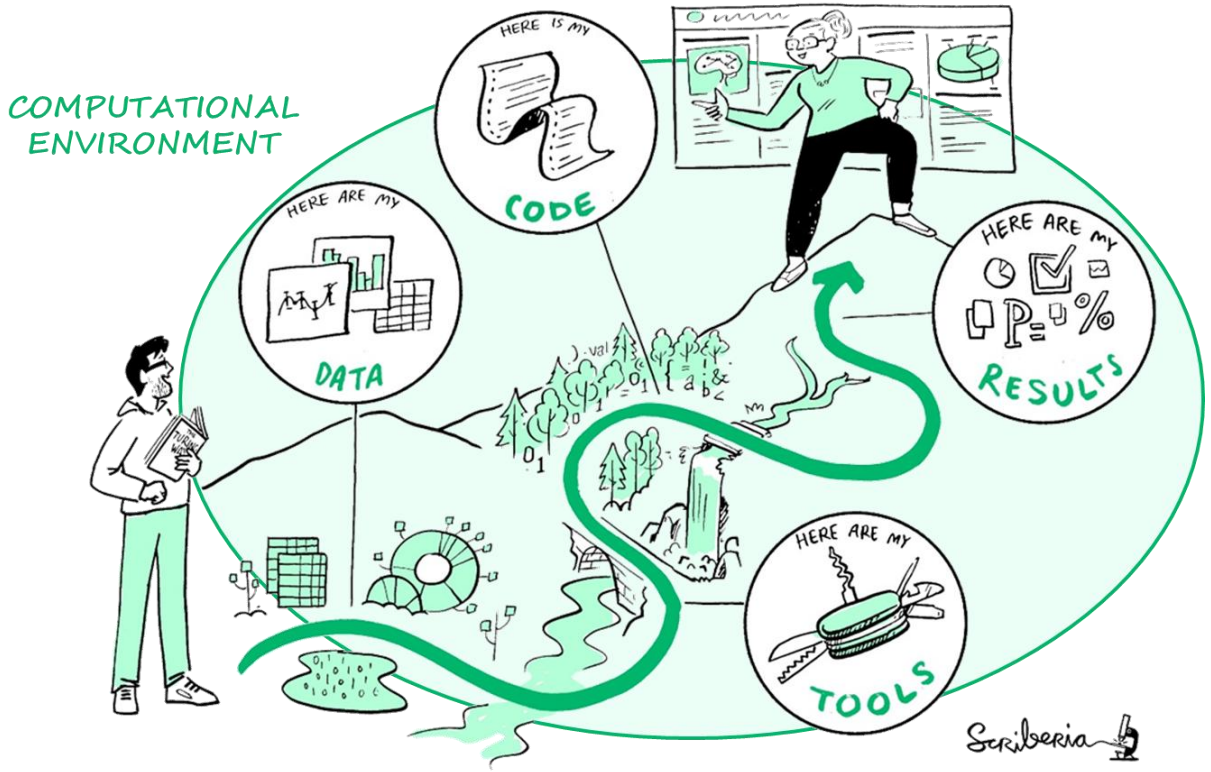
Agenda

Quarto @ Swiss TPH
Introduction to Quarto
Structure of a Quarto document
Quarto key features
Swiss TPH templates
Q&A (and/or a break)
Hands-on session

Introduction to Quarto

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Reproducibility in research



Same data
+Same tools
+Same code
=Same results

This illustration is adapted from the one created by Scriberia with The Turing Way community, and used under a CC-BY 4.0 licence. DOI: 10.5281/zenodo.3332807.

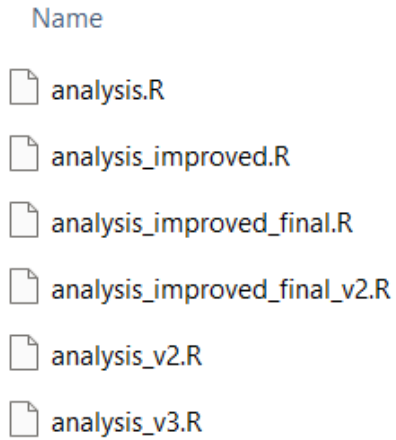
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Imagine you sent your collaborator a report

...and after six months

‘Could you please exclude those who under 18 and regenerate Figure 3?’

And your project folder looks like:

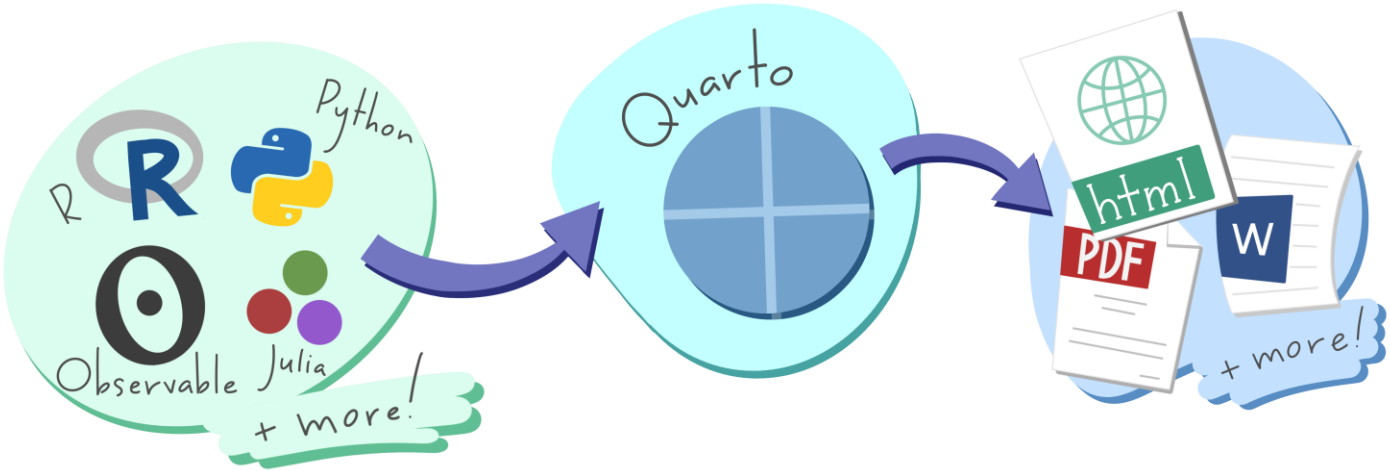


You may struggle to find out which script you used...

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What is Quarto

An open-source scientific and technical publishing system.



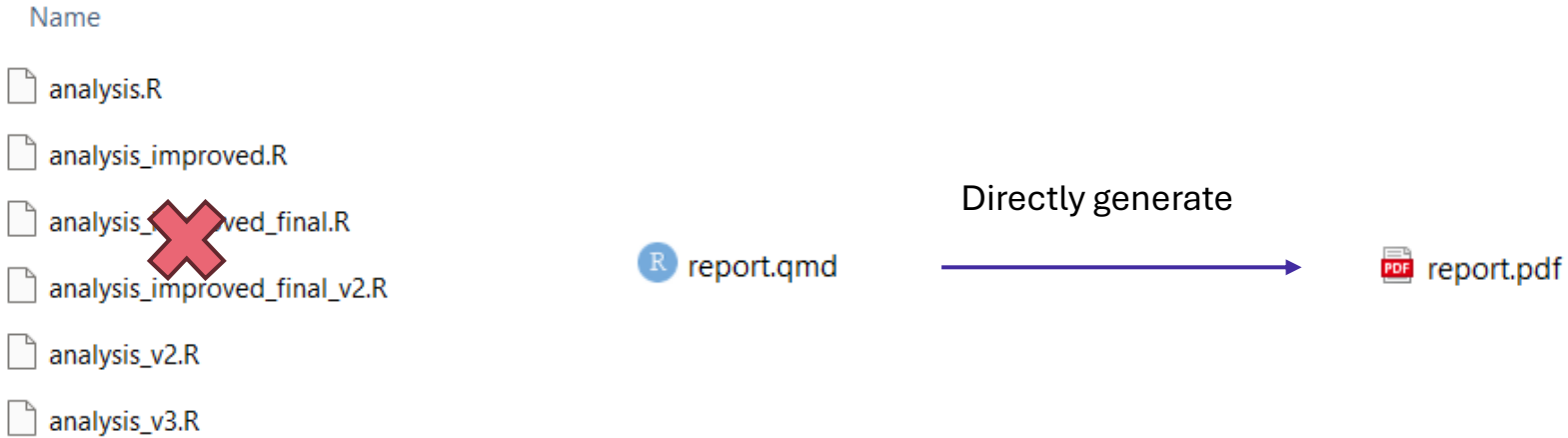
 Artwork by Allison Horst.

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One source of truth

Quarto has everything in one place

Your narrative, your code, and your results should live in **one document** and be generated in **one step**.



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Bonus: A better practice

Source file + version control







A Quarto document:
The source of truth all in one file

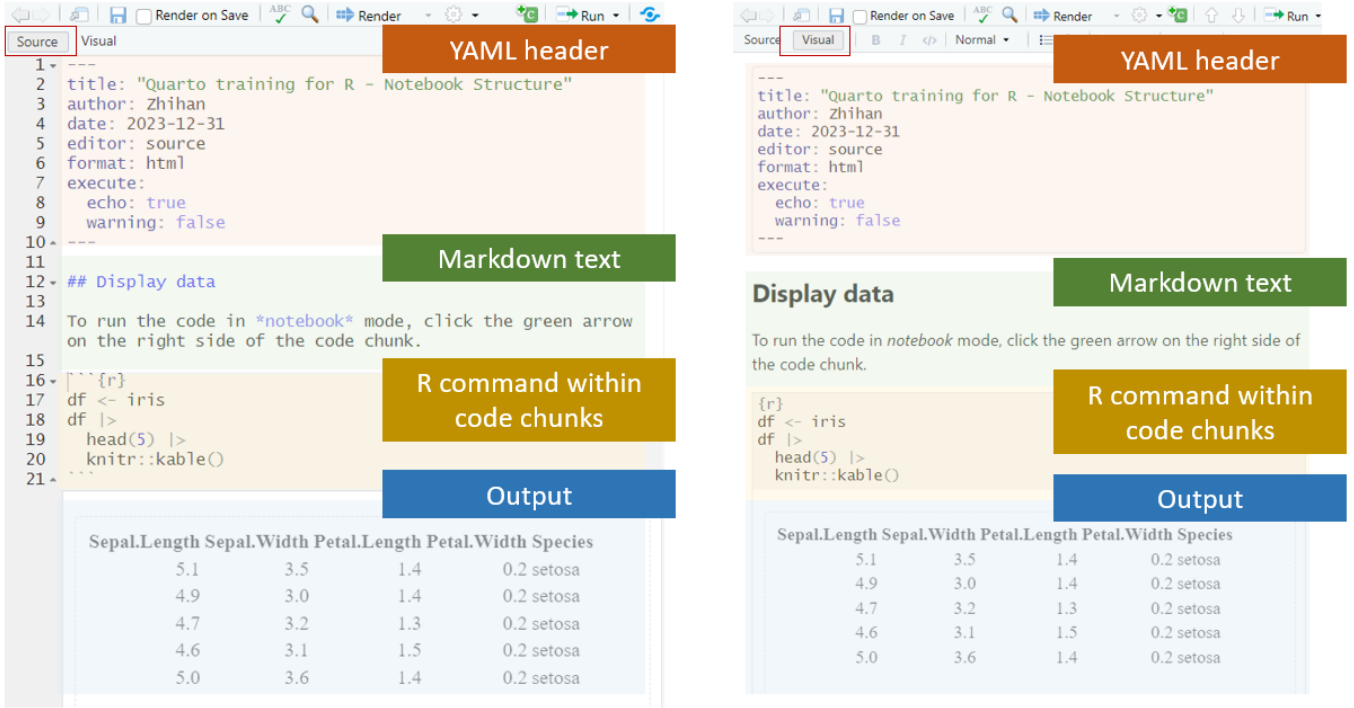
Version control tool:
Different versions over time

Structure of a Quarto document

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Structure of Quarto document

source and visual editor



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Structure of Quarto document

YAML header

- Contains commands and metadata.
- One header at the top of the document.
- Surrounded by `---` on the top and bottom of the section.

```yaml
---
title: "TITLE OF DOCUMENT"
additional1: "ADDITIONAL LINE 1"
additional2: "ADDITIONAL LINE 2"
subtitle: "Subtitle of Document"
projectname: "Name of Project/Mandate, etc..."
author:
  - name: Jane Doe
    affiliation: University of Basel
    email: jane.doe@unibas.ch
  - name: Sam Smith
    affiliation: Swiss TPH
    email: sam.smith@swisstph.ch
format:
  report-typst:
    section-numbering: "1.1.1"
numbering: 1
date: last-modified
fontsize: 11pt
lang: en
execute:
  echo: false
  warning: false
toc: true
toc-depth: 3
bibliography: references.bib
---
```

Structure of Quarto document

Markdown text

- A markup language, which can have formatting.
- This is where you have your free text.

```
186  # TIPS AND TRICKS
187
188  ## Swiss TPH self-paced Quarto tutorial
189
190  See [Swiss TPH Quarto training](https://research-it-swiss-tph.github.io/quarto_training/) for the Quarto tutorial website
     developed by Research Informatics.
```



3. TIPS AND TRICKS

3.1 Swiss TPH self-paced Quarto tutorial

See Swiss TPH Quarto training for the Quarto tutorial website developed by Research-IT.

Element	Markdown Syntax

Heading	`# H1`
	`## H2`
	`### H3`

Bold	`**bold text**`
Italic	`*italicized text*`
Blockquote	`> blockquote`

Ordered List	`1. First item`
	`2. Second item`
	`3. Third item`

Unordered List	`- First item`
	`- Second item`
	`- Third item`

Code	`` `code` ``
Horizontal Rule	`---`
Link	`[title](https://www.example.com)`
Image	`![alt text](image.jpg)`

Markdown cheatsheet

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Structure of Quarto document

Code blocks

- This is where you write your executable commands, e.g. R.
- They are identified by backticks + language name surrounded by braces, e.g. {r}
- You can have cell level options at the top of the code block.

````
```{r}
#| label: tbl-flextable
#| tbl-cap: "Car Performance Data (mtcars)"
#| tbl-colwidths: auto

# Load required packages
library(flextable)
library(dplyr)

# Prepare data (first 5 rows of mtcars)
data <- head(mtcars, 5) %>%
  tibble::rownames_to_column("Car Model") %>%
  select(1:4) # Select first 4 columns for brevity

# Create flextable
ft <- flextable(data) %>%
  # Style headers
  bold(part = "header") %>%
  bg(part = "header", bg = "#432CA1") %>%
  color(part = "header", color = "white") %>%
  # Style body
  bg(bg = "white", part = "body") %>%
  # Column-specific formatting
  colformat_num(j = c("mpg", "disp"), digits = 1) %>%
  # Autofit columns
  autofit()

ft
```
````


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Structure of Quarto document

Output

- Output of your code can be displayed inline in your IDE and/or in the rendered document.

````
```{r}
#| label: tbl-flextable
#| tbl-cap: "Car Performance Data (mtcars)"
#| tbl-colwidths: auto

# Load required packages
library(flextable)
library(dplyr)

# Prepare data (first 5 rows of mtcars)
data <- head(mtcars, 5) %>%
  tibble::rownames_to_column("Car Model") %>%
  select(1:4) # Select first 4 columns for brevity

# Create flextable
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  # Style headers
  bold(part = "header") %>%
  bg(part = "header", bg = "#432CA1") %>%
  color(part = "header", color = "white") %>%
  # Style body
  bg(bg = "white", part = "body") %>%
  # Column-specific formatting
  colformat_num(j = c("mpg", "disp"), digits = 1) %>%
  # Autofit columns
  autofit()

ft
```
````

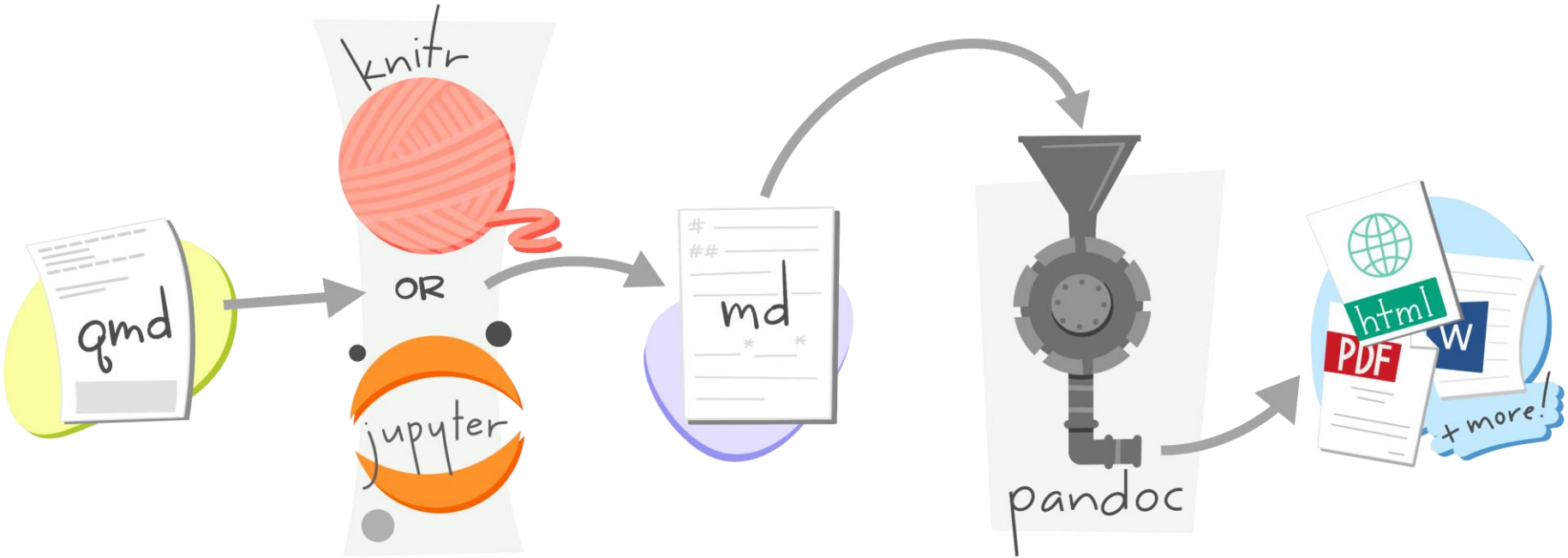
240ms Just now

Car Model	mpg	cyl	disp
Mazda RX4	21.0	6	160
Mazda RX4 Wag	21.0	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

Quarto key features

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Multi-format output

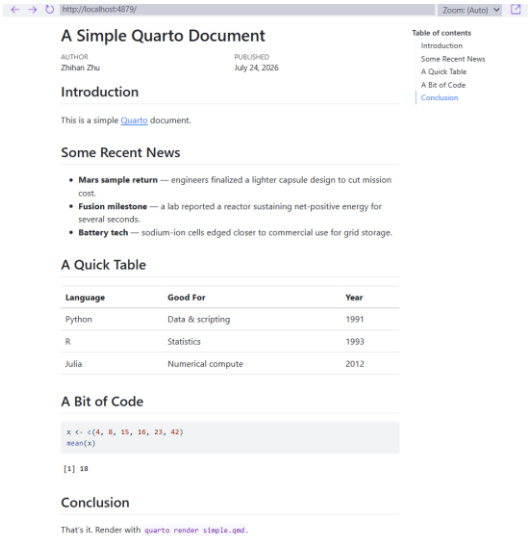


 Artwork by Allison Horst.

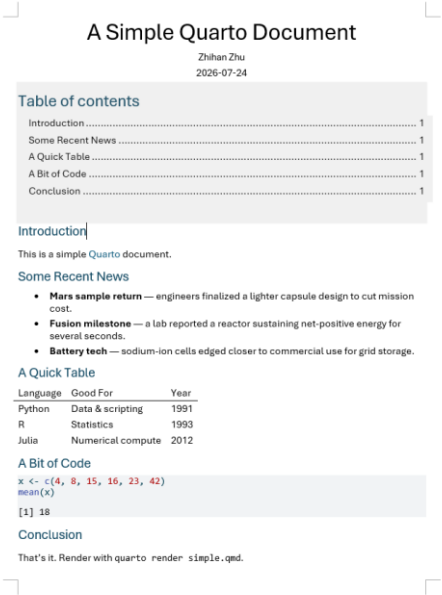
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Control output format in YAML header

`format: html`

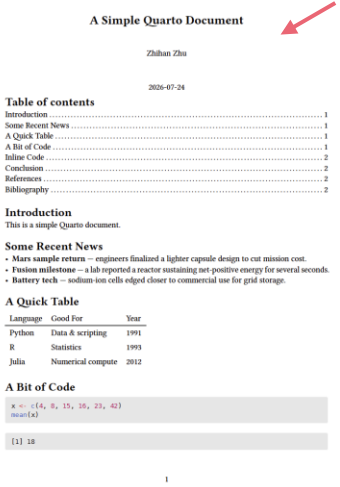


`format: docx`



`format: typst`

This is a PDF file



 More options can be found in Guide – Quarto.

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Font and font size

An example

Source Visual

```
1  ---
2  title: "A Simple Quarto Document"
3  author: "Zhihan Zhu"
4  date: 2026-07-24
5  format: typst
6  mainfont: "Times New Roman"
7  fontsize: 12pt
8  toc: true
9  bibliography: references.bib
10 ---
```

Alert: The font settings may be different in different output formats – please always refer to the official documentations!

Output options

Option	Description
`eval`	Evaluate the code chunk (if `false`, just echos the code into the output).
`echo`	Include the source code in output
`output`	Include the results of executing the code in the output (`true`, `false`, or `asis` to indicate that the output is raw markdown and should not have any of Quarto's standard enclosing markdown).
`warning`	Include warnings in the output.
`error`	Include errors in the output (note that this implies that errors executing code will not halt processing of the document).
`include`	Catch all for preventing any output (code or results) from being included (e.g. `include: false` suppresses all output from the code block).
`renderings`	Specify rendering names for the plot or table outputs of the cell, e.g. `[light, dark]`

More options can be found in Guide – Quarto.

Output options

echo: show the code or not?

```
30 ## A Bit of Code
31
32 ```{r}
33 #| echo: true
34 x ← c(4, 8, 15, 16, 23, 42)
35 mean(x)
36 ```
```

echo: true

A Bit of Code

```
x <- c(4, 8, 15, 16, 23, 42)
mean(x)
```

```
[1] 18
```

echo: false

A Bit of Code

```
[1] 18
```


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Output options

output: show the results or not?

```
30 ## A Bit of Code
31
32 ```{r}
33 #| echo: true
34 x ← c(4, 8, 15, 16, 23, 42)
35 mean(x)
36 ```
```

output: true

A Bit of Code

```
x <- c(4, 8, 15, 16, 23, 42)
mean(x)
```

```
[1] 18
```

output: false

A Bit of Code

```
x <- c(4, 8, 15, 16, 23, 42)
mean(x)
```

Output options

include: run the code, but show it or not (the master off-switch)

include: true

A Bit of Code

```
x <- c(4, 8, 15, 16, 23, 42)
mean(x)
```

```
[1] 18
```

include: false

A Bit of Code

Where to use the options

````
---
title: "Knitr Document"
execute:
  echo: false
---

```{r}
#| warning: false

library(ggplot2)
ggplot(airquality, aes(Temp, Ozone)) +
  geom_point() +
  geom_smooth(method = "loess", se = FALSE)
```

```{r}
summary(airquality)
```
````

Can be in the YAML header to become global options

Or in a code block to become local options

Inline code

Execute code within markdown

```
## Inline Code

The mean of `x` is `r mean(x)`.
```



Inline Code

The mean of x is 18.

Cross-reference

````
```{r}
#| label: tbl-flextable
#| tbl-cap: "Car Performance Data (mtcars)"
#| tbl-colwidths: auto

# Load required packages
library(flextable)
library(dplyr)

# Prepare data (first 5 rows of mtcars)
data <- head(mtcars, 5) %>%
  tibble::rownames_to_column("Car Model") %>%
  select(1:4) # Select first 4 columns for brevity

# Create flextable
ft <- flextable(data) %>%
  # Style headers
  bold(part = "header") %>%
  bg(part = "header", bg = "#432CA1") %>%
  color(part = "header", color = "white") %>%
  # Style body
  bg(bg = "white", part = "body") %>%
  # Column-specific formatting
  colformat_num(j = c("mpg", "disp"), digits = 1) %>%
  # Autofit columns
  autofit()

ft
```


@tbl-flextable is an example of using R code to generate a table in Quarto documents.
````

Refer to a specific table or figure

Tips:

- In order for a figure to be cross-referenceable, its label must start with the *fig-* prefix.
- In order for a table to be cross-referenceable, its label must start with the *tbl-* prefix.

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Cross-reference

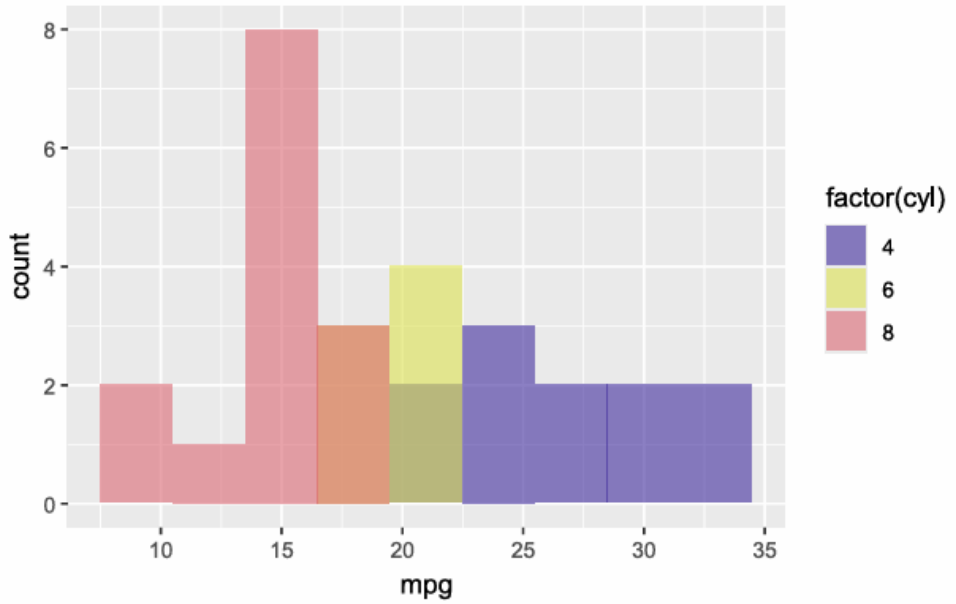


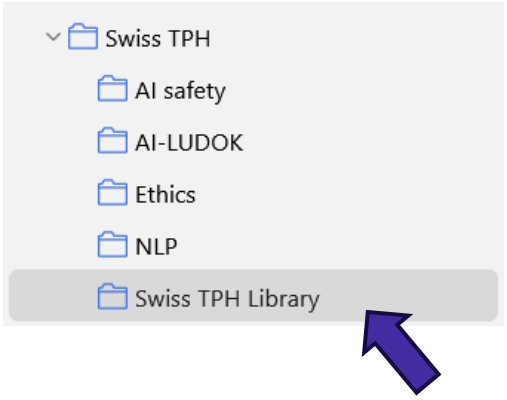
Figure 1: Miles Per Gallon (MPG) Distribution

Figure 1 is an example of using R code to generate a figure in Quarto documents.

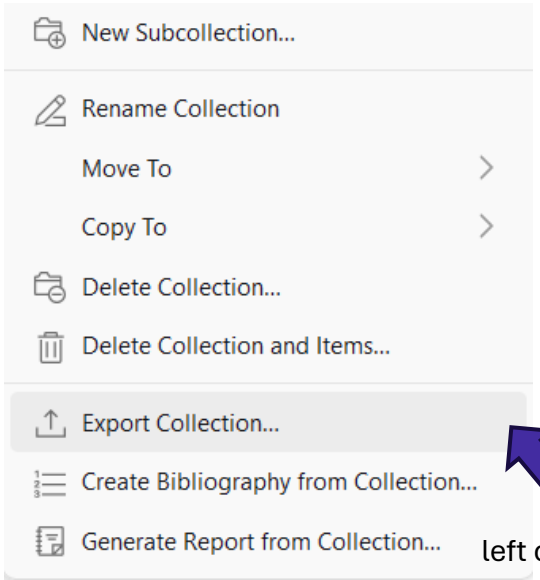
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Citations and bibliography

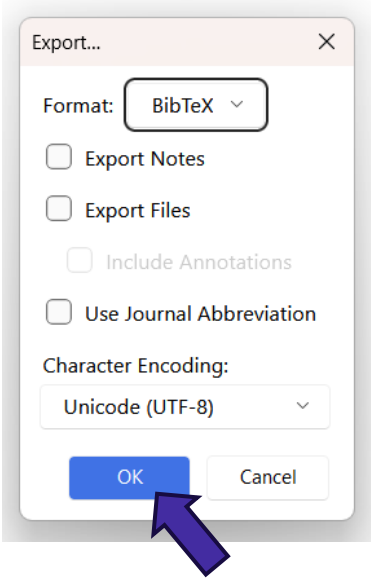
How to export .bib file? (Zotero example)



right click



left click



It's recommended to put .bib file under the same directory of .qmd file!

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Citations and bibliography

```yaml
---
title: "A Simple Quarto Document"
author: "Zhihan Zhu"
date: 2026-07-24
format: pdf
toc: true
bibliography: references.bib
---
```

Add bibliography in the YAML header

```
## Conclusion

That's it. Render with `quarto render simple.qmd`.

For more on R and data analysis, see @wickham2023r4ds.

## References

::: {#refs}
:::
```

Cite it in your text

Print the reference at the end of the document

Change citation style: Citations – Quarto

Citations and bibliography

What it looks like in the rendered file

```
40  ## Conclusion
41
42  That's it. Render with `quarto render simple.qmd`.
43
44  For more on R and data analysis, see @wickham2023r4ds.
45
46  ## References
47
48  ::: {#refs}
49  :::
```



Conclusion

That's it. Render with `quarto render simple.qmd`.

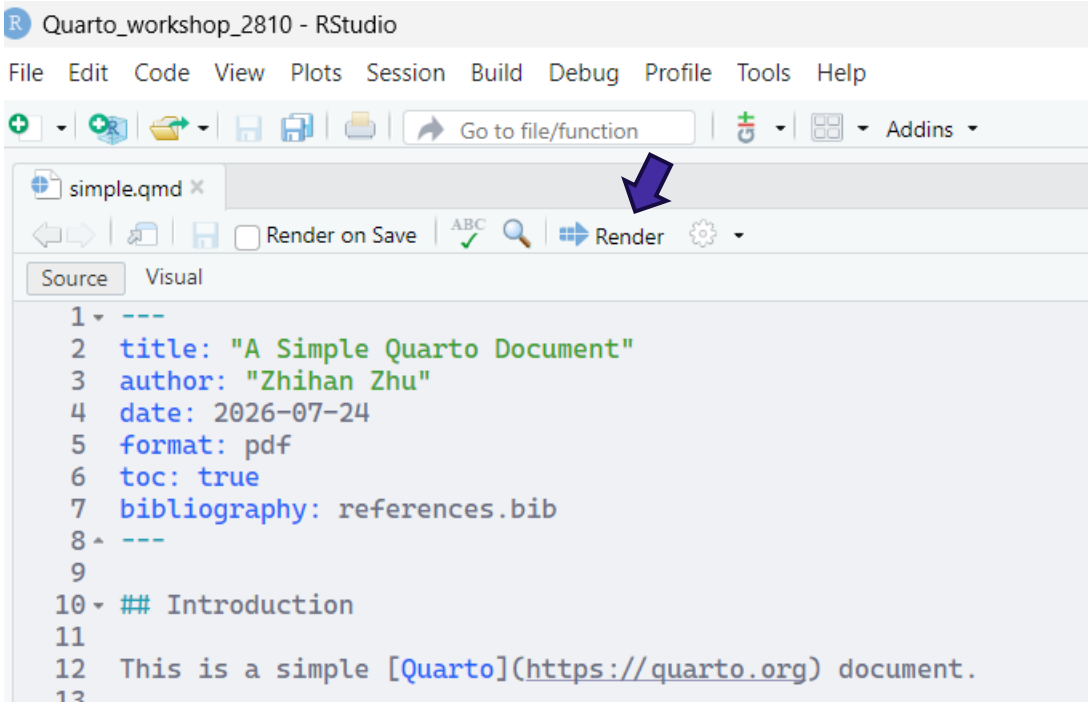
For more on R and data analysis, see Wickham, Çetinkaya-Rundel, and Grolemund (2023).

References

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science*. 2nd ed. Sebastopol, CA: O'Reilly Media.

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How to render Quarto documents in RStudio



Successful render depends on correct syntax of all components (YAML header, Markdown text, Code...)

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Bonus: Parametrised reports

A demo

Task: You are going to prepare population reports for all cantons in Switzerland.

Challenge:

- Each canton has its own data, so
- *Without Quarto*: You might need to create 26 separate Word documents, generate every single plot, copy paste them one by one into the documents. The details (counts, names, etc.) in the documents need to be edit by hand.
- *With Quarto*: You can create 26 reports with a single click using parametrised reports.

Bonus: Extensions

Existing public extensions: Quarto Extensions – Quarto

And you can create your personal extensions…

Swiss TPH Templates

Swiss TPH

Swiss TPH templates

What are available?

For now, Swiss TPH Quarto templates are available in these formats:

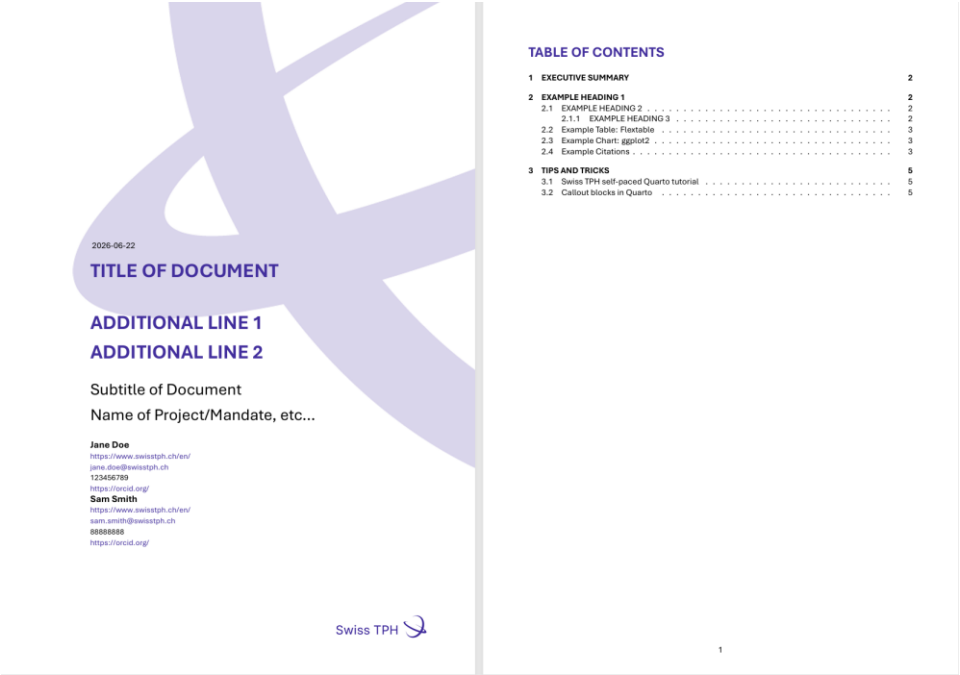
- PDF report (Latex and Typst)
- PDF letterhead (Latex and Typst)
- Word
- Powerpoint
- Revealjs

Contributors: Hélène Langet, Zhihan Zhu

Research-IT-Swiss-TPH/stphtoolbox: Swiss TPH R toolbox for reproducible research

Swiss TPH Quarto template gallery

PDF(Latex) cover page and table of content



2026-06-22

TITLE OF DOCUMENT

ADDITIONAL LINE 1
ADDITIONAL LINE 2

Subtitle of Document

Name of Project/Mandate, etc...

Jane Doe
https://www.swisstph.ch/en/
jane.doe@swisstph.ch
12345789
https://orcid.org/
Sam Smith
https://www.swisstph.ch/en/
sam.smith@swisstph.ch
88888888
https://orcid.org/

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1

Swiss TPH

Swiss TPH Quarto template gallery

PDF(Latex) headings, text, tables and plots

1. EXECUTIVE SUMMARY

Dolor euismod primis etiam cubilia interdum, nec: mauris, ut nunc consequat tempor parturient odio? Purus elementum mauris curae – urna nullam aenean ultrices molestie nisi. Accumsan fusce enim cubilia tortor erat sem suscipit fusce. Ridiculus dui, in sociis nullam, molestie ad natoque tempor. Parturient aliquam quisque pulvinar penatibus accumsan, pharetra sollicitudin ac netus. Rutrum eleifend aenean dignissim nibh nullam lacinia torquent lectus aptent donec? Class neque inceptos pharetra pulvinar.

Amet luctus, taciti ornare sociis inceptos nunc nisi eleifend! Sollicitudin fringilla nec enim sociis mi augue felis maecenas dictumst. Habitant dapibus lobortis nulla platea, magnis praesent vestibulum orci. Justo dapibus varius, a, scelerisque quis fermentum semper erat enim! Enim nam torquent, cum pellentesque dictumst. Lobortis bibendum odio aliquam volutpat erat ut ornare. Ornare dictum ultrices montes aliquam egestas morbi – aliquam magna blandit dis venenatis: auctor primis tempor elementum id feugiat taciti nascetur cum molestie.

2. EXAMPLE HEADING 1

Sit non pretium suspendisse metus pellentesque nisl at sociosqu. Dui venenatis gravida ornare diam, nulla sollicitudin etiam nisl mattis parturient? Rhoncus sed auctor facilisis arcu, platea mauris – duis ultricies ultrices? Curae et dictum mi, nec placerat faucibus cubilia varius sodales! Porttitor molestie dignissim mollis dui est, mus; netus suscipit vitae rhoncus.

2.1 EXAMPLE HEADING 2

Dolor ullamcorper nunc penatibus nulla sociosqu integer: mauris diam non. Nibh mauris, ac varius risus laoreet quisque dictum sed etiam velit rutrum? Euismod convallis a platea quam imperdiet faucibus varius potenti gravida inceptos. Venenatis sodales ultrices massa fringilla sodales risus curae viverra. Viverra accumsan morbi sociis!

Amet rutrum vel magna faucibus neque torquent; risus vel massa, lectus nunc! Phasellus nostra felis fringilla curae litora nunc tincidunt blandit netus cras habitasse maecenas arcu. Scelerisque hac netus a at nullam curabitur; lacus porttitor. Ligula lobortis odio ridiculus nec mi cum cubilia – litora nulla class accumsan. Feugiat sollicitudin arcu magnis: rhoncus sociosqu torquent est. Venenatis phasellus viverra taciti mollis tortor!

2.1.1 EXAMPLE HEADING 3

Amet auctor ultricies orci dictumst felis: ultrices proin aliquet at: leo neque sem duis. Sed metus tincidunt gravida pharetra commodo, eget ultrices ultricies! Sodales ante consequat tincidunt nunc nam et magnis. Nascetur hac inceptos venenatis auctor accumsan porta condimentum volutpat ridiculus? Pellentesque diam sodales, facilisis ultrices conubia maecenas. Potenti luctus eleifend cubilia venenatis torquent – ultrices convallis quam. Na quam.

Ipsum senectus phasellus sodales eget aptent hendrerit congue conubia nostra nascetur penatibus. Rhoncus in vivamus pellentesque iaculis molestie lobortis urna. Ultrices tempus tempor, fringilla donec augue nascetur elementum vestibulum. Molestie rutrum mus nunc curae nam, iaculis libero pretium fusce. Cursus dictumst nisl tortor, pellentesque ut.

EXAMPLE HEADING 4

Elit orci ullamcorper non rhoncus cum cubilia id dui ante senectus. Ut senectus litora mus cum sed feugiat, vitae imperdiet urna. Donec eget duis malesuada quam dapibus ridiculus eros eros: auctor odio risus.

2.2 Example Table: Flextable

Table 1: Car Performance Data (mtcars)

Car Model	mpg	cyl	disp
Mazda RX4	21.0	6	160
Mazda RX4 Wag	21.0	6	160
Datsun 710	22.8	4	108
Hornet 4 Drive	21.4	6	258
Hornet Sportabout	18.7	8	360

Table 1 is an example of using R code to generate a table in Quarto documents.

2.3 Example Chart: ggplot2

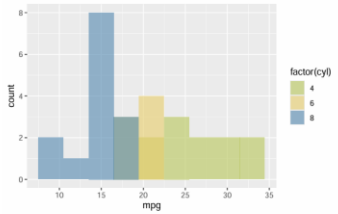


Figure 1: Miles Per Gallon (MPG) Distribution

Figure 1 is an example of using R code to generate a figure in Quarto documents.

2.4 Example Citations

This example is from Akpodiete et al. (2025).

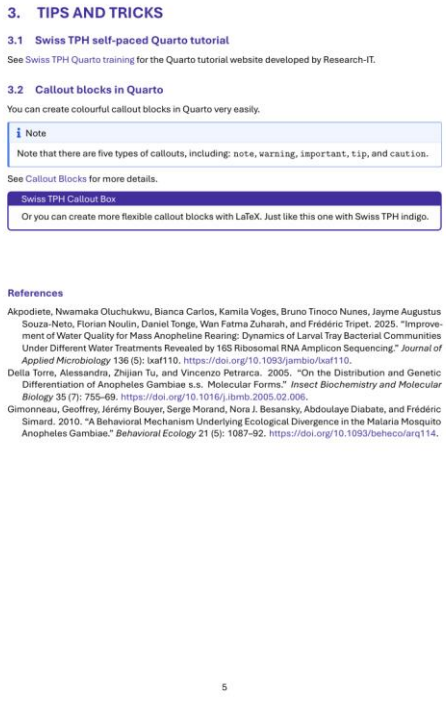
Mosquitoes of the *Anopheles gambiae* s.l. complex are the primary vectors of human malaria in sub-Saharan Africa. The freshwater species *Anopheles coluzzii* is one of this complex's most abundant and widespread vectors. It is found from Senegal to Angola, spanning West Africa, West-Central Africa, and North-West Southern Africa (Della Torre, Tu, and Petrarca (2005), Gimonneau et al. (2010))

For more advanced use of Quarto citations, see Citations - Quarto.

Swiss TPH

Swiss TPH Quarto template gallery

PDF(Latex) callout boxes, references, and more...



3. TIPS AND TRICKS

3.1 Swiss TPH self-paced Quarto tutorial

See Swiss TPH Quarto training for the Quarto tutorial website developed by Research-IT.

3.2 Callout blocks in Quarto

You can create colourful callout blocks in Quarto very easily.

Note

Note that there are five types of callouts, including: `note`, `warning`, `important`, `tip`, and `caution`.

See Callout Blocks for more details.

Swiss TPH Callout Box

Or you can create more flexible callout blocks with LaTeX. Just like this one with Swiss TPH indigo.

References

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Swiss TPH

Q&A (or take a break)

Swiss TPH

Hands-on

Swiss TPH

Hands-on instructions

1. For those who did not receive the GitHub link, come to me or go to https://github.com/Research-IT-Swiss-TPH/quarto-workshop-2026
2. Download course material.
3. Go to exercises folder, open report-starter.qmd and report-instructions.pdf.
4. Follow the instructions in report-instructions.pdf.
5. Edit everything in report-starter.qmd according to the instructions

Thank you for your attention

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Swiss TPH

Resources

- Quarto official guide: Guide – Quarto
- Swiss TPH tools & Quarto templates GitHub page: Research-IT-Swiss-TPH/stphtoolbox: Swiss TPH R toolbox for reproducible research
- Swiss TPH Quarto self-learning: Swiss TPH Quarto training – Quarto training